Darin Lei

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EDUCATION

University of Toronto **Toronto, ON**
Honours Bachelor of Science Expected 2027
Psychology Research Specialist (Thesis Program); Cognitive Science Major
Thesis: Determining the mechanisms of temporal context recovery in autobiographical memory
Advisor: Dr. Morgan D. Barense
cGPA: 3.89/4.00

AWARDS & ACADEMIC HONOURS

John Davidson Ketchum Memorial Scholarship in Psychology (CA$1,230) 2026
Best Undergraduate Poster Presentation Award, Ontario Student Neuroscience Conference 2026
University of Toronto Excellence Award (CA$7,500) 2026
Dean's List Scholar of the Faculty of Arts & Science 2025–2026
DSC International School Award for Excellence in Language Scholarship (HK$2,000) 2023
Ontario Scholar Award 2023

MANUSCRIPTS IN PREPARATION

Abdulrazeq, H., Sapirstein, E., **Lei, D.**, & Lega, B. (2026). *Neuroplasticity and cognitive recovery after temporal lobe epilepsy surgery: A systematic review with lessons from stroke, traumatic brain injury, and primary progressive aphasia rehabilitation.*

STUDENT-REVIEWED PUBLICATIONS

Lei, D. (in press). Learning by uncertainty: Children's help-seeking as a developmental apparatus for adaptive, dynamic problem-solving. *Inkblot: The Undergraduate Journal of Psychology*, *13*.

Lei, D., & Djedovic, A. (2025). The Synergistic Mind-Body (SAIS) Hypothesis: A Daoist approach to consciousness from the perspective of traditional Chinese medicine. *Canadian Undergraduate Journal of Cognitive Science*, *8*, e7230. https://doi.org/10.21810/cujcs.v8i1.7230

Lei, D., & Vervaeke, J. (2026). From 'Ha Ha!' to 'Aha!': Towards an integrated understanding of the effect of humour comprehension on insight problem-solving. *Mentis: Journal of Cognitive Science and Artificial Intelligence, 1*(1), 40–56.

PRESENTATIONS

Lei, D., Lohnas, L., Seger, S., Lega, B., & Barense, M.D. (2027, March 20-23). *Context as an anchor of episodic recollection: A test of the context maintenance and retrieval (CMR) framework in the real world* [Poster presentation]. The Cognitive Neuroscience Society (CNS) 34th Annual Meeting, Boston, MA, USA.

Lei, D., Barense, M. D., Seger, S., Lohnas, L., & Lega, B. (2026, August 24). *Determining the mechanisms of temporal context recovery in autobiographical memory with ambulatory intracranial recording* **[Awarded Best Poster Presentation].** Ontario Student Neuroscience Conference (OSNC), Toronto, ON, Canada.

Lei, D., Hong, L., & Campos, J. L. (2026, August 13). *Unravelling the relationship between mental health, vection, and hearing loss: A lifespan approach* [Poster presentation]. University Health Network Summer Training and Research (STAR) Program Research Day, Toronto, ON, Canada.

Lei, D., Brachaniec, M., Hong, B., Hughes, E. C., & Barense, M. D. (2026, June 2). *The HippoCamera Knowledge Hub Program* [Conference presentation]. AGE-WELL Early Professionals, Inspired Careers

in AgeTech 7th Annual Conference (EPIC-AT), Toronto, ON, Canada.

Lei, D., Hughes, E. C., Hong, B., & Barense, M. D. (2026, February 5–6). *The HippoCamera Knowledge Hub: Developing a digital educational program for older adults to promote healthy aging* [Poster presentation]. Lake Ontario Visionary Establishment 53rd Annual Conference (LOVE), The Brock Niagara Falls Fallsview, Niagara Falls, ON, Canada.

Hong, B., **Lei, D.**, Hughes, E. C., & Barense, M. D. (2025, October 22–23). *Developing a digital educational program for older adults to improve memory: A person-based approach* [Conference presentation]. AGE-WELL 10th Annual Conference (AGEWELL2025), Palais des Congrès, Montréal, QC, Canada.

Lei, D., Hong, B., Hughes, E. C., & Barense, M. D. (2025, September 25). *The HippoCamera Educational Program: Developing a digital educational program to improve memory in older adults: A person-based approach* [Poster presentation]. University of Toronto St. George Campus Research Opportunity Program (ROP) Research Poster Fair, Toronto, ON, Canada.

Lei, D., & Vervaeke, J. (2025, April 8*). The "High" Mind: Towards a unified theoretical account of insight problem-solving, predictive processing, and relevance realization through oneirogenic psychedelics* [Poster presentation]. University of Toronto St. George Campus Psychology Undergraduate Research Community (PURC) 2nd Annual Undergraduate Symposium, Toronto, ON, Canada.

RESEARCH EXPERIENCE

Texas Computational Memory Lab, University of Texas Southwestern Medical Center
PI: *Dr. Bradley Lega* **Dallas, TX and Toronto, ON (Hybrid)**
External Research Collaborator January 2026—Present

- Conduct literature reviews on recovery outcomes for memory and language in patients with temporal lobe epilepsy (TLE) post-resection, especially as they relate to plasticity.
- Will administer the Autobiographical Interview (AI; Levine, 2002) with TLE patients implanted with the NeuroPace Responsive Neurostimulation (RNS) system.
- Will analyse the temporal recall dynamics of autobiographical memory data by using multivariate and linear mixed models in RStudio.
- Training in localizing electrodes in anatomical brain scans of TLE patients following RNS surgical implant procedures in FreeSurfer.
- Design and implement patient tutorials into the HippoCamera smartphone application to guide RNS data syncing to the PDMS, supporting data collection in epilepsy patients.

Levine Lab, Rotman Research Institute, Baycrest Academy for Research and Education
PI: Dr. Brian Levine **Toronto, ON**
Research Assistant January 2026—Present

- Conduct literature reviews on the memory functions of sleep (specifically with respect to the formation of episodic memories in autobiographical memory), targeted memory reactivation (TMR), and the use of naturalistic stimuli in memory research paradigms.
- Contribute to an ongoing meta-analysis of studies employing the Autobiographical Interview (Levine et al., 2002) as an approach to quantifying episodic richness in memory recall.
- Attend weekly lab meetings discussing ongoing research on human memory with a focus on individual differences in autobiographical memory and traumatic memories.
- Present ongoing work in my assigned project at lab meetings.
- **Main Project:** Investigating the role of sleep in episodic memory consolidation
    - Over a century of research from the cognitive and clinical neuroscientific literatures has consistently demonstrated an integral role of sleep in episodic memory consolidation and recollection. Although robust, much of the evidence supporting this finding comes from laboratory-based paradigms involving recognition tasks for non-naturalistic stimuli such as wordlists. As such, it remains unclear whether sleep benefits the free recall of episodic

memories and what neurophysiological mechanisms might be at play in such an effect. Here, we seek to examine the impact of sleep on free recall by using a naturalistic memory design in the form of a controlled hospital tour. Ultimately, we aim to identify the role of sleep by investigating the effect of targeted memory reactivation (TMR) on episodic recall. To our knowledge, this study is the first to investigate the role of sleep for free recall under the context of naturalistic memory.

Memory & Perception Lab, Department of Psychology, University of Toronto
PI: Dr. Morgan D. Barense **Toronto, ON**
Study Lead/Thesis Student May 2025—Present

- Conduct literature reviews on autobiographical memory and computational models of human memory (e.g., Context Maintenance and Retrieval; CMR), cognitive decline in older adults, behavioural change frameworks, and interventions for age-related memory decline.
- Develop an educational program on the HippoCamera app where older adults can learn more about research-backed strategies to improve their memory through designing prototypes on Figma.
- Evaluate the usability and acceptability of the educational program by consulting older adults in the Greater Toronto Area in semi-structured interviews.
- Perform quantitative data analyses through RStudio and reflexive thematic analysis using NVivo.
- Attend weekly lab meetings discussing ongoing research in human memory.
- Present ongoing work in my assigned projects at lab meetings.
- Collect autobiographical memory data from healthy young adult participants using HippoCamera.
- Interview prospective research assistants.
- **Thesis Project:** Determining the mechanisms of temporal context recovery in autobiographical memory (in collaboration with Dr. Bradley Lega, University of Texas Southwestern Medical Center)
    - Episodic recollection depends on recovering not only the content of an event, but also its temporal context. Retrieved-context theories predict that recalling an episode reinstates its encoding context and thereby biases recovery toward temporally adjacent events. However, this key prediction has only been tested in highly controlled, lab-based experiments that lack the rich personal context of autobiographical memory. This project uses HippoCamera to test retrieved-context predictions in real-world autobiographical memory, first behaviourally in healthy adults (data collection nearing completion), and then in epilepsy patients implanted with the NeuroPace RNS system to examine the neural mechanisms. To our knowledge, this is the first study to directly test retrieved-context mechanisms in real-world autobiographical memory.
- **Individual Research Project:** Developing technology-based solutions to promote healthy aging
    - Aging is associated with physical and cognitive changes that can greatly affect the autonomy and well-being of older adults. This project aims to explore the efficacy of technology-based solutions to address the unique needs of older adults through the development of an educational program on a digital memory augmentation app called HippoCamera. By implementing this program, older adults can learn how to apply different memory strategies in their daily life with benefits for memory retention.

Cognitive Science Program, University of Toronto
Supervisor: Dr. John Vervaeke **Toronto, ON**
Independent Study Project Student October 2026—Present

- Develop a theoretical account of consciousness, informed by published information-theoretic work using partial information decomposition (PID), with a focus on the role of psychedelics, humour, anaesthesia, and insight problem-solving, extending prior work on predictive processing and relevance realization.

- Conduct literature reviews on the relationship between insight problem-solving and humour, neuropsychopharmacology of psychedelics and anaesthetic agents, and theories of consciousness.
- **Main Project:** Consciousness as a global workspace of recursive relevance realization as revealed by insight problem-solving, humour, anaesthesia, and psychedelics
    - The emergence of consciousness is a cognitive phenomenon that remains enigmatic and largely unsolved. However, converging findings from cognitive science, psychiatry, and the neurosciences point towards a few clues that may aid in the development of an integrative theory of consciousness. These clues come in the form of humour, insight problem-solving, psychedelics, and anaesthesia. Although these phenomena are often treated independently, they all reveal pivotal insights into the nature and functional properties of consciousness. This project seeks to explore the hypothesis that consciousness can be understood through these four phenomena, leveraging recent theoretical frameworks to do so.

Multisensory Integration in Virtual Environments (MIVE) Lab, KITE Research Institute, Toronto Rehabilitation Institute, University Health Network
PI: Dr. Jennifer L. Campos **Toronto, ON**
Individual Summer Research Project Student May—August 2026

- Conducted literature reviews on the relationships between hearing loss and vection and vestibular dysfunction, compensatory strategies for regaining balance in individuals with hearing loss.
- Attended weekly lab meetings on a wide range of topics including vection, hearing loss, technological innovations in geriatric care, and visually induced motion sickness (VIMS).
- Analysed electromyography (EMG) and posturography data.
- Presented ongoing work in my assigned project at lab meetings.
- Administered psychometric assessments including Dynamic Visual Acuity (DVA), Trail-Making Test, Stroop, Pure Tone Test, Canadian Digit Triplet Task (CDTT), Pelli-Robson Contrast Sensitivity Chart, Randot Stereo Test, MoCA, and Sound Induced Flash Illusion (SIFI), and a Rod and Disk Task to obtain baseline measures of cognitive, visual, auditory, and vestibular functions.
- **Main Project:** Visual perturbation in vection and hearing loss
    - It is well established that individuals with hearing loss, especially older adults, tend to struggle with balance. To compensate for these deficits in balance, individuals may react either by stiffening their bodies or moving dynamically with the direction of motion. However, these compensatory strategies are diametrically opposed in what they suggest about the sensorimotor integration and vestibular function as a whole. This project seeks to determine not only what strategies are used in hearing loss to combat issues with balance but also the mechanisms that underlie such strategies, and their resultant effects on illusory perception of self-motion (vection).

LEADERSHIP EXPERIENCE

University of Toronto Karate Club, Club Sports, Sport & Rec, Faculty of Kinesiology & Physical Education, University of Toronto **Toronto, ON**
Club President May 2026—Present
Vice President June 2025—May 2026

- Organize inter-university events such as the Inter-University Karate Tournament, which provides over 100 athletes across Ontario with a post-secondary competitive environment.
- Liaise with University of Toronto Faculty of Kinesiology and Physical Education staff to arrange facilities access and training spaces for club athletes.
- Manage club administration, including fees, training, grading, attendance, and relations with the Ontario Karate Federation and other clubs across Ontario (e.g., Waterloo, Queen's, Toronto Metropolitan University).

- Attend biweekly meetings on club outreach and instructional and seminar-based lesson design.
- Lead various social activities within the club and manage the team's online presence.

Shorinji-Kan Jiu Jitsu Club, Fitness and Wellness, Hart House, University of Toronto **Toronto, ON**
Assistant Instructor October 2025—August 2026

- Assisted with the instruction of beginner and intermediate classes.
- Demonstrated core techniques and supervised technical drills involving takedowns, throws, joint manipulations, and breakfalling.
- Provided helpful feedback to support skill acquisition with an emphasis on control and safety.
- Maintained a safe and supportive learning environment by assisting with first aid in cases of injury.
- Helped manage warm-ups, class structure, training flow, and cooldowns with the Head Instructor.

Freelance Academic Tutoring Services **Hong Kong, China (Remote) & Toronto, ON**
High School English, Math & Science Tutor May 2022—January 2025

- Taught struggling students senior-level high school courses in English, Calculus & Biology.
- Created additional resources in the form of lectures, worksheets, annotations, citation guides, writing guides, essay assignments, and mock examinations related to the Ontario Secondary School Literacy Test (OSSLT) and the IELTS.

VOLUNTEER EXPERIENCE

University Health Network STEM Pathways, University Health Network **Toronto, ON**
Youth Ambassador and Volunteer May 2026—August 2026

- Supported early education and career development in youth (Grades K–12) from underserved and underrepresented communities by assisting with the organization of STEM education and outreach events such as interactive tours, networking panels, hands-on workshops, and classroom visits.
- Gave tours of the Challenging Environments and Assessment Laboratory (CEAL) at the Toronto Rehabilitation Institute to high school students.
- Instilled a passion and curiosity for STEM by leading scientific workshops and participating as a panellist in career networking sessions.

Psychology Graduate Students Association, Department of Psychology, Faculty of Arts & Science, University of Toronto **Toronto, ON**
Academic Poster Judge May 2026

- Evaluated four academic poster presentations by graduate students enrolled with the Department of Psychology based on criteria of academic rigour, delivery, and presentation style established by the Chair of an academic psychology conference.

University of Toronto Psychedelics Journal Club **Toronto, ON**
Discussion Leader & Presenter December 2024—Present

- Led discussions on recent publications in the psychedelic literature on the utility of psychedelic substances in therapy and for the exploration of altered states of consciousness as they pertain to frameworks in cognitive science.
- Presented notable findings (of choice) in the literature, offering a wide-ranging scope of the theoretical basis of such findings and their implications for the empirical and practical uses of psychedelic substances, e.g., in research and in the clinic.

CERTIFICATIONS

WHMIS Training, University Health Network	2026
Safe Sport Training, Coaching Association of Canada	2025
TCPS2 CORE 2022	2025
MathWorks Core MATLAB Skills, Computer Vision Onramp, & Deep Learning Onramp	2025

Certificate of Completion, Montreal Cognitive Assessment (MoCA)	2025
CIHR Institute of Gender and Health Core Competency Modules for Sex and Gender in Biomedical Research, Primary Data Collection with Human Participants, and the Analysis of Secondary Data from Human Participants	2025
The Johns Hopkins University, Fundamental Neuroscience for Neuroimaging Course	2024
The University of Hong Kong, Academy for the Talented, Junior Neuroscience Program	2021

RELEVANT COURSEWORK

PSY359 Human Neuroimaging Laboratory (A+)	PSY380 Vision Science (A)
PSY312 Cognitive Development (A+)	PSY396 Neurochemical Basis of Behaviour (A)
PSY309 Research Specialization: Practicum (A+)	COG341 Attention, Perception, Consciousness (A)

TECHNICAL SKILLS

Programming & Data Analysis	R, Python, MATLAB, JavaScript, HTML5/CSS, jamovi, JASP, NVivo (qualitative analysis), Taguette
Neuroimaging Software	FSL (FEAT/FLIRT/BET/MELODIC/TBSS/FSLEYES), FreeSurfer
Experiment Design & Programming	PsychoPy, Psychtoolbox (MATLAB), Qualtrics, Lookit, REDCap
Technical Equipment	EyeLink 1000 Plus, EyeLink DataViewer, EDF2ASC
Databases & Literature Review Tools	Zotero, PubMed, PsycINFO, Google Scholar, Excel, Covidence
Design & Graphics	Adobe Photoshop, Adobe Lightroom, Figma, Canva, Blender
User App Testing	TestFlight (user testing as a UX researcher)

LANGUAGES

Native	English, Mandarin
Fluent	Heritage Cantonese, Heritage Taiwanese Hokkien
Conversational	French, Japanese